

University Websites as Orientation Infrastructures: A Systematic Audit of Navigability, Discoverability, Accessibility and Information Exposure in Italian Universities

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Abstract—Italian university websites act as central orientation infrastructure for prospective students, yet their effectiveness depends on the joint behaviour of distinct construct layers usually studied in isolation. This paper reports a systematic, multi-construct audit of twenty Italian universities and forty-one degree programmes, combining (i) homepage-to-course-page journey audits, (ii) automated technical and accessibility evaluation with Lighthouse and WAVE, (iii) external discoverability through controlled search-engine queries, and (iv) structural information extraction from course pages. The four layers are integrated into a validated audit matrix yielding construct-level interpretive flags. We find that high external findability coexists with persistent internal friction, recurrent accessibility violations, and uneven information exposure, and that critical cases concentrate around shared technological infrastructure rather than individual institutions. The contribution is both methodological and empirical: a replicable audit framework that integrates the four constructs while keeping them distinct rather than collapsing them into composite scores.

Index Terms—university websites, orientation infrastructure, web accessibility, Web Content Accessibility Guidelines (WCAG) 2.1, discoverability, Italian higher education, audit methodology

I. INTRODUCTION

For prospective students, choosing a university degree is a high-stakes informational decision: candidates compare programmes, admission rules, fees, deadlines and curricular structures across institutions before committing for several years. Promotional videos, open days, brochures and social media support this comparison, but programme-specific information is primarily stabilised and updated through institutional web pages [1], [2]. Once on the institutional domain, students must reach the relevant course page; once there, the page must expose decision-relevant data. Both steps are infrastructural:

take place in digital environments that universities provision, and students' navigation may occur without prior knowledge of internal terminology.

Existing research tends to evaluate these interfaces one construct at a time. The dominant strand targets Web Content Accessibility Guidelines (WCAG) conformance [3]–[7], a pattern confirmed by recent bibliometric mapping [8]. Studies assessing usability from a prospective-student perspective, focusing on admission pages, enrolment tasks or decision-oriented navigation, are present but form a smaller subset [1], [9], [10]. No prior work combines, on the same sample, homepage-to-course-page journeys, automated technical and accessibility evaluation, external SERP (search-engine results page) discoverability, structural information extraction from course pages, and an explicit reading of the multi-platform ecosystem in which those courses live.

This paper addresses this gap through a multi-construct audit of Italian universities, where a binding accessibility framework has long applied to public-sector digital services [11]–[13]. We treat the website as an *orientation infrastructure* [14] and measure its behaviour along four analytically separable constructs: navigability, technical and accessibility quality, external discoverability, and structural information exposure.

We pose five research questions. RQ1: to what extent does the institutional website guide the prospective student from the homepage to a specific course page? RQ2: how does external search engine behaviour interact with internal navigation? RQ3: do the resulting pages meet automated technical and accessibility quality thresholds? RQ4: do the pages expose the information needed to support a decision? RQ5: which combinations of failure modes recur across the sample, and

how are they distributed across institutions?

The contribution is twofold. Empirically, we report a comparative, descriptive picture of forty-one programmes across twenty Italian universities, covering all four constructs on the same sample. Methodologically, we contribute a replicable audit protocol that keeps the four constructs distinct rather than collapsing them into composite scores, and release the validated analytical layer over which interpretive flags are computed.

The paper is organised as follows. Section II contextualises the work in research on university websites, accessibility, prospective-student orientation and the Italian higher-education web ecosystem. Section III describes the methodology, explaining the sample, audit framework, provenance model and tool limitations. Section IV reports the descriptive results by research question and integrates them across constructs. Section V discusses the implications for orientation infrastructure, accessibility governance and shared web platforms. Section VI provides a conclusion.

II. RELATED WORK AND CONTEXT

A. University websites as object of study

Web accessibility and usability of higher-education websites are established research areas, dominated by automated audits and cross-country comparisons [4], [5], [15], [16]. Systematic reviews report that, across regions and institution types, most university websites do not fully meet WCAG [3], [4], [7]. In Italy, Barricelli *et al.* repeatedly find persistent non-conformance with national accessibility requirements [13]. However, this literature largely treats pages or homepages as the unit of analysis, leaving navigation paths and external discoverability less systematically modelled.

B. Accessibility, WCAG, and the Stanca Act

The Web Content Accessibility Guidelines 2.1 [3] are the de facto reference standard. In Italy, public-sector websites, including those of public universities, are subject to the Stanca Act [11] and AgID accessibility guidelines [17], which require WCAG 2.1 Level AA conformance [18]. Empirical evidence shows that legal mandates do not ensure substantive conformance: in Finland, where comparable obligations apply to higher education, all 38 audited landing pages contained accessibility errors [6]; in Italy, public-university websites still display widespread violations [13]. Similar patterns are reported in Latin America [15] and the Gulf region [5], [16].

C. Prospective students and orientation

A smaller body of work focuses on prospective students. Macakoğlu *et al.* [1] compare prospective-student pages across Europe, North America and Oceania along usability, accessibility and security; Diwanji [2] examines inclusiveness for international students. Logue *et al.* [14] explicitly frame transfer-information pages as *orientation infrastructure*, documenting how inconsistent information across units of the same system hinders mobility. Ghorbanzadeh and Sharbatian [19] link website features to value co-creation and brand reputation.

These studies suggest that the website is not a brochure but an active layer of student experience. However, real navigation, external findability and structural exposure remain only partially connected, rather than integrated on a shared sample.

D. Italian higher-education web ecosystem

A specific feature of the Italian context, rarely explicit in the literature, is that the institutional website functions as a multi-platform ecosystem. Several layers coexist: an institutional content management system (CMS); CINECA-based student-management and course-catalogue services, including the widely used ESSE3 [20]; annual ANVUR programme documentation (SUA-CdS) [21]; national ministerial portals; Moodle [22] as the dominant learning management system (LMS); and departmental sub-domains. AgID design guidelines [17] prescribe a coherent public-sector design language but do not eliminate platform-level fragmentation. This shared backbone is methodologically relevant: page-level results may partly reflect vendor-level decisions rather than individual institutional choices, and the same course often has multiple legitimate public faces.

III. METHODS

A. Sample

The sample comprises twenty Italian universities and forty-one programmes in communication, media and neighbouring areas: twenty first-cycle, twenty second-cycle and one single-cycle programme. Table I lists all institutions and programmes; Table II reports measurement coverage by phase. Institutions were selected to span the national territory and to vary in size, institutional type and postgraduate offer.

B. Audit framework

The audit comprises five phases.

a) *Phase 1 – Journey audit (RQ1)*: For each programme, we run a structured navigation from the institutional homepage to course page, recording entry point, click depth, elapsed time, dominant path archetype (*programmes_hub*, *menu*, *on-site_search*, *cta*, *mixed*), and use of internal search and filters. Journeys are tagged *success* when the canonical course page is reached, and *blocked_js* when the institutional path depends on client-side rendering that does not resolve under the audited session conditions (anonymous desktop browser, Italian interface, 1440×900 viewport, no prior login); the latter is treated as a valid finding, not as missing data. We record support evidence (HTML snapshots, screenshots) for each journey.

b) *Phase 2 – Lighthouse and WAVE (RQ3)*: On the twenty homepages and the forty-one course pages reached in Phase 1 (sixty-one targets), we collect Google Lighthouse [23] audits in mobile strategy and WAVE [24] accessibility evaluations. For two course pages whose Angular front-end prevented the WAVE HTTP API from returning results, evaluation was completed manually using the WAVE browser extension for Google Chrome.

TABLE I
SAMPLE: INSTITUTIONS AND PROGRAMMES (N = 41)

Institution	Area	Programme	Cycle
Unibo	N	Scienze della comunicazione	L
		Semiotica	LM
Unipd	N	Comunicazione	L
		Strategie di comunicazione	LM
UniPi	C	Discipline dello spettacolo e della comunicazione	L
		Comunicazione, Media, Tecnologie	LM
UniFi	C	Scienze Umanistiche per la Comunicazione	L
		Pratiche, linguaggi e culture della comunicazione	LM
UniSi	C	Scienze della comunicazione	L
		Strategie e tecniche della comunicazione	LM
UniTo	N	Scienze della comunicazione	L
		Comunicazione e culture dei media	LM
UniGe	N	Media, comunicazione e società	L
		Digital Humanities – Interactive systems and digital media	LM
UniMiB	N	Comunicazione interculturale	L
		Teoria e tecnologia della comunicazione	LM
UniMi	N	Comunicazione e società	L
		Editoria, culture della comunicazione e della moda	LM
Insubria	N	Scienze della comunicazione	L
		Scienze e Tecniche della Comunicazione	LM
Unisa	S	Scienze della comunicazione	L
		Corporate Communication e Media (interclasse LM-59/LM-92)	LM
Roma Tre	C	Scienze della comunicazione	L
		Scienze cognitive della comunicazione e dell'azione	LM
UniPa	I	Scienze della Comunicazione	L
		Comunicazione del patrimonio culturale	LM
UniStraPG	C	Comunicazione internazionale e pubblicitaria	L
		Comunicazione pubblicitaria, storytelling e cultura d'immagine	LM
UniUrb	C	Informazione, Media, Pubblicità	L
		Comunicazione Digitale per la Cultura	LM
Uninettuno	O	Comunicazione digitale d'impresa	L
		Culture, linguaggi e strategie per la comunicazione nell'era digitale	LM
UniCal	S	Comunicazione e DAMS	L
		Linguaggi della comunicazione e dello spettacolo	LM
UniNa	S	Culture digitali e della comunicazione	L
		Digital Society, Social Innovation and Global Citizenship	LM
UniBg	N	Scienze della Comunicazione	L
		Valorizzazione del patrimonio culturale materiale e immateriale	LM
UniTrento	N	Beni culturali	L
		Organizzazione, società e tecnologia	LM
		Giurisprudenza	LM-CU

Area: N = North (9 inst.), C = Centre (6), S = South (3), I = Islands (1), O = Online (1). Cycle: L = first, LM = second, LM-CU = single.

TABLE II
MEASUREMENT COVERAGE

Measurement	n	Notes
Journey audits	41	Homepage to course page
Technical targets	61	20 homepages + 41 course pages
SERP observations	123	3 controlled queries per programme
Structural records	41	18 indicators per course page

All four constructs achieve complete coverage; 61 technical targets = 20 homepages + 41 course pages.

c) *Phase 3 – SERP discoverability (RQ2)*: For each programme, we issue three controlled queries – a degree-and-institution canonical query (SQ01), a topic-driven query (SQ02), and an exploratory keyword query (SQ03) – and record the rank and match type of the institutional target in the top-10 search results. The match policy distinguishes exact matches to the canonical course URL, normalized canonical matches (e.g. protocol, trailing slash, anchors or tracking parameters), official equivalent surfaces for the same course (e.g. departmental pages, course-platform pages, LMS surfaces

or CINECA-hosted catalogues), related but non-equivalent official pages, third-party results, ambiguous cases and not-found observations. Only canonical, normalized and official-equivalent results are counted as target-found; the other cases are retained as contextual evidence but not as successful findability.

d) *Phase 4 – Structural extraction (RQ4)*: Each course page is scored against eighteen structural indicators, partitioned into ten *essential*, five *important*, and three *contextual* indicators. Essential indicators are course title, degree level, academic year, location, language, study plan, admission requirements, admission procedure, deadlines and contacts. Important indicators are degree class, credits, duration, career outcomes and official regulation. Contextual indicators are fees or costs, accessibility services and quality or satisfaction information. Detection combines raw HTTP and headless-browser rendering when JavaScript is required.

e) *Phase 5 – Audit matrix (RQ5)*: The four phases are joined into three analytical tables at course, institution and page level. The course-level table contains identifiers, journey outcomes, SERP findings, Lighthouse and WAVE metrics, structural indicators and construct-level flags. The institution-

TABLE III
TECHNICAL AND ACCESSIBILITY QUALITY: SUMMARY STATISTICS
(N = 61 TARGETS)

Metric	Mean	Median	Min	Max
Lighthouse accessibility	0.93	0.94	0.78	1.00
Lighthouse mobile perf.	0.58	0.59	0.05	0.97
WAVE errors	7.9	4	0	126
WAVE alerts	76.4	44	1	507
WAVE contrast errors	10.9	3	0	76

Lighthouse accessibility high on average (0.93); WAVE error counts reveal a heavy right tail; mobile performance substantially lower (mean 0.58).

level table aggregates the same signals by university, while the page-level table preserves homepage and course-page technical measures separately. A controlled missingness vocabulary is applied to all cells; thresholds for interpretive flags are either distributional (typically $p75$) or tool-based (Lighthouse category breakpoints) and are documented in the released dataset.

C. Provenance and reproducibility

Each reported value is traceable to a dataset, a phase-level report and an evidence bundle (page captures, SERP screenshots, raw tool outputs) preserved per crawl run.

IV. RESULTS

A. Internal navigability (RQ1)

All forty-one journeys were completed under the protocol. Thirty-five terminated with *success*; six were classified *blocked_js*, meaning that the institutional path did not resolve to the target course page under the audited browsing conditions. These non-resolved journeys were concentrated in four institutional contexts: one programme at Bologna (*Semiotica*), two at Padova (*Comunicazione*; *Strategie di comunicazione*), two at Napoli Federico II (*Culture digitali e della comunicazione*; *Digital Society*, *Social Innovation and Global Citizenship*), and one at Trento (*Organizzazione, società e tecnologia*). On successful journeys the mean click depth is 2.43 (median 2) and the mean elapsed time is 27.3 s (median 24 s). The dominant archetype is *programmes_hub*, followed by *menu* and *onsite_search*. Internal search is used in nine journeys, filters in seven, and a direct CTA shortcut in five.

B. Technical and accessibility quality (RQ3)

Lighthouse and WAVE were collected on all sixty-one targets (Table III). Lighthouse accessibility scores cluster near the upper bound (mean 0.93), but WAVE error and contrast counts reveal a long right tail that the aggregate score tends to mask: the worst-performing page records 126 WAVE errors and 76 contrast failures. Mobile Lighthouse performance is substantially lower (mean 0.58, range 0.05–0.97), indicating uneven investment in mobile optimisation across the sample.

C. External discoverability (RQ2)

Of 123 SERP observations, 119 (96.7%) returned the official target or an official equivalent in the top-10 results; 102 of these had organic rank one and 119 had rank in the top three.

The four *not_found* cases are all on SQ01, the exact degree-and-institution query. Match types are 80 *canonical_exact*, 31 *official_equivalent*, 8 *canonical_normalized* and 4 *not_found*. The recurrence of *official_equivalent* matches confirms that the same programme has multiple legitimate public surfaces – including departmental pages, LMS/course-platform pages and CINECA-hosted catalogues – competing with the canonical course page. Observer notes for the SQ01 misses also record third-party aggregators, including AlmaLaurea and Masterin.it, where the official target did not surface.

D. Structural information exposure (RQ4)

Across the forty-one course pages, three essential indicators – course title, degree level and contacts – are present at full coverage; study-plan information is present in 39 of 41 cases; deadlines, by contrast, are present in only 22 of 41, the lowest-coverage essential indicator (Table IV). Nine course pages cover all ten essential indicators, while ten cover six or fewer. Eight pages required headless-browser rendering for structural detection, indicating reliance on client-side composition for content delivery. The pattern is therefore not a generic absence of course information: identification fields are robust, while decision-critical operational fields such as deadlines, language, admission requirements and location are unevenly exposed.

TABLE IV
COVERAGE OF ESSENTIAL STRUCTURAL INDICATORS (N=41)

Indicator	Present	Not observed
Course title	41	0
Degree level	41	0
Contacts	41	0
Study plan	39	2
Admission procedure	30	11
Admission requirements	29	12
Location	29	12
Academic year	27	14
Language	24	17
Deadlines	22	19

Identification fields (title, level, contacts) at full coverage; deadlines absent from 46% of pages.

E. Cross-layer integration (RQ5)

Cross-layer integration is performed by preserving each construct as a separate diagnostic signal. Eleven programs are classified as high-friction journeys, which means that a successful path still requires at least four clicks, at least 35 seconds, or an internal search. Eight of these eleven cases reach the course page through an internal search. Twelve programmes cross at least one accessibility-risk threshold (WAVE errors, WAVE contrast errors or Lighthouse accessibility), and eleven cross the technical-risk threshold for mobile Lighthouse performance. Only one programme is marked as information-fragmented, and no programme is weakly findable externally because all programmes appear in at least one top-10 SERP result across the controlled queries.

At institution level (Table V), the risks are not evenly distributed. Bologna, Padova, Calabria and Napoli Federico II each contribute two technical-risk programmes; Pisa, Torino,

TABLE V
INSTITUTION-LEVEL CROSS-LAYER PROFILE. COUNTS ARE PROGRAMMES PER INSTITUTION; WEAK EXTERNAL FINDABILITY IS OMITTED BECAUSE IT IS 0 FOR ALL INSTITUTIONS.

Institution	Courses	Blocked	High friction	Accessibility risk	Technical risk	Essential coverage
Unibo	2	1	1	0	2	86%
Unipd	2	2	0	0	2	57%
UniPi	2	0	2	2	0	86%
UniFi	2	0	2	0	0	93%
UniSi	2	0	1	0	0	79%
UniTo	2	0	2	2	0	93%
UniGe	2	0	0	2	0	93%
UniMiB	2	0	2	0	0	79%
UniMi	2	0	0	0	0	100%
Insubria	2	0	0	0	0	100%
Unisa	2	0	0	0	0	71%
Roma Tre	2	0	0	0	1	100%
UniPa	2	0	0	2	0	86%
UniStraPG	2	0	0	0	0	86%
UniUrb	2	0	0	0	1	100%
Uninetuno	2	0	0	0	1	71%
UniCal	2	0	0	1	2	64%
UniNa	2	2	0	2	2	64%
UniBg	2	0	1	1	0	86%
UniTrento	3	1	0	0	0	95%

Napoli Federico II is the only institution with blocked journeys on both programmes; risk distribution follows shared platform choices more than institution size or type.

Genova, Palermo and Napoli Federico II each contribute two accessibility-risk programmes. The two multi-phase critical cases are *Culture digitali e della comunicazione* and *Digital Society, Social Innovation and Global Citizenship*, both at Napoli Federico II. They combine a blocked journey, accessibility risk and technical risk on the same Angular single-page-application front-end.

V. DISCUSSION

A. Internal friction and successful journeys

A journey's success may conceal the cognitive cost it imposes, such as with the search box. Eight of the eleven high-friction cases use the on-site search box as a recovery mechanism when direct navigation fails, leading to a mischaracterized positive affordance. The blocked-JavaScript cases are a stronger version of this concealing: a page that exists on the server but does not exist for a standard browser is, for the purposes of orientation, missing.

B. External findability and internal fragmentation

External search engines surface the institutional target, or an official equivalent, in 96.7% of observations. This high external findability is, however, distributed across multiple legitimate official surfaces (institutional course page, departmental page, LMS/course-platform page, CINECA-hosted catalogue), each with potentially different contents and update schedules. The exact degree-and-institution query is also the only query type with *not_found* observations, and the observer notes in those cases point to third-party aggregators rather than the canonical course page. From the point of view of the prospective student, the bottleneck shifts from finding the program to reconciling its multiple public representations

C. Accessibility law compliance

Mean Lighthouse accessibility scores near 0.93 are a reassuring aggregation, but obscure a heavy right tail in WAVE

error and contrast counts. Twelve of forty-one course pages cross at least one of the three accessibility-risk thresholds. The pattern is consistent with Finland's experience under a comparable legal mandate [6] and with the Italian fifteen-year retrospective [13], showing that a binding law does not produce conformance without continuous auditing and remediation.

D. Critical cases cluster around a shared infrastructure

The two multi-phase critical programmes, *Culture digitali e della comunicazione* and *Digital Society, Social Innovation and Global Citizenship* at Napoli Federico II, share a single front-end stack. They are blocked in the internal journey and also cross both technical and accessibility risk thresholds. This is consistent with the multi-platform ecosystem framing: page-level results can carry a platform-level signature. Audit findings should therefore be read as evidence on the ecosystem and on the local choice of stack, not solely on individual editorial work.

E. Information exposure on essential indicators

The course page reliably carries identification data (title, degree level, contacts), but is more variable on actionable data, such as deadlines. As deadlines are a decision-critical field for prospective students, their absence or burial penalizes mobility and benefits well-connected applicants who can recover the information through informal channels. This is a tractable point of intervention with high informational return.

F. Limitations

As our findings are point-in-time snapshots on a purposive sample, further future longitudinal monitoring will be needed to monitor the persistency of the identified issues. Tool-wise, Lighthouse and WAVE can automatically detect only a subset of WCAG criteria: as systematic reviews consistently confirm [4], [7], expert and user-based evaluation remain a

necessary complement to automated diagnostic signals. Additionally, SERP observations were collected under controlled conditions (incognito browsing, Italian locale), but SERP results are inherently dynamic and partially personalised; rank estimates near the top-10 boundary therefore carry residual uncertainty that controlled queries can reduce, but not eliminate. Finally, Phase 1 is observer-mediated and documents plausible prospective-student navigation rather than an exhaustive enumeration of possible paths. The structured protocol and supporting evidence bundle reduces, but do not eliminate residual inter-rater variance.

These limitations can impact the precision of individual findings, without undermining the construct-level patterns, which are consistent across the sample and across the independent measurement instruments.

VI. CONCLUSION

University websites in the Italian system function as orientation infrastructure for prospective students, and their quality depends on four jointly necessary constructs: internal navigability, external discoverability, technical and accessibility quality, and structural information exposure. The findings reveal a clear asymmetry: high external findability coexists with internal friction, persistent accessibility violations and uneven exposure of decision-critical fields. Critically, the most severe multi-layer failures concentrate around shared platform choices rather than individual institutions, making the audit a diagnostic of the ecosystem, not merely of editorial work. The contribution is methodological as well as empirical: a replicable audit protocol that keeps constructs distinct and releases validated analytical tables. Future work should extend to the full Italian university population, add disability-targeted user studies and pursue longitudinal monitoring.

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